TEA: A Training-Efficient Adapting Framework for Tuning Foundation Models in Remote Sensing

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***Abstract*— With well-pretrained foundation models (FMs), the performance of almost every remote sensing interpretation task has been boosted. The parameter volume of FMs increases with their continuously enhanced capabilities, leading to increased costs of fine-tuning. To apply FMs more effectively and efficiently, there are already some arts that introduce the parameter-efficient fine-tuning (PEFT) concept into remote sensing and achieve competitive performance with much lower parameter cost. However, the training efficiency of most PEFT frameworks may be not satisfactory. To make tuning FMs for remote sensing applications more efficient, we propose a training-efficient adapting (TEA) framework. Specifically, we attach a SIDE adapter network (SIDEAN) to the frozen powerful FMs and only update the SIDEAN to perform the downstream tasks. Moreover, to make TEA perceive remote sensing scenes from a macroscopic perspective and boost the performance, we propose a top-down guidance mechanism to inject macro scene information into the SIDEAN during adapting. TEA is also parameter-efficient, as SIDEAN is designed to be lightweight. We conduct extensive experiments to demonstrate the effectiveness and efficiency of TEA on ten widely adopted datasets covering four primary remote sensing tasks, e.g., object detection, orientated object detection, semantic segmentation, and scene classification. By training only 5.43% of the frozen FM parameters, TEA can save more than 57% of training memory footprint and up to 15% of time cost on average while achieving competitive performance on all datasets. Furthermore, TEA can surpass full fine-tuning on several datasets.**

***Index Terms*— Foundation model (FM), parameter-efficient fine-tuning (PEFT), remote sensing.**

I. Introduction

DEEP learning-based remote sensing image interpretation methods have achieved remarkable progress [1], [2], [3], [4] and are applied in wide fields, such as agriculture [5], [6], urban planning [7], [8], [9], and environmental monitoring [10], [11]. Lately, with the advancement of pretraining techniques [12], [13], [14], [15], [16], pretrained foundation models (FMs) have become a research hotspot in remote sensing community. FMs can leverage massive remote sensing data and significantly enhance the performance of downstream tasks [12], [13], [17], [18], [19], [20]. With the thriving development of FMs, the parameter volume of FMs increases dramatically alongside their capabilities. [21], [22]. Applying a large FM by the standard full fine-tuning protocol for downstream tasks becomes more training- and storage-consuming, as full fine-tuning requires updating and saving an entire copy of FM for each downstream task. For example, the parameter volume of Swin-L is 195M which is about 8.3× larger than that of the commonly used ResNet50 in the past, and the training memory and time are about 4.1× and 2.3× larger, respectively. How to reduce the fine-tuning cost of FMs for remote sensing downstream tasks while keeping impressive performance emerges as a necessary research topic.

Parameter-efficient fine-tuning (PEFT) is commonly studied to optimize the resource consumption of fine-tuning. Researchers try to reduce the trainable parameters per task to alleviate the storage issue while keeping comparable performance [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34]. Many novel works insert lightweight trainable blocks or tokens into pretrained FMs and freeze the pretrained parameters [25], [26], [27], [28], [29], [30], [31], while some turn to tune a selected subset of pretrained parameters [32], [33], [34]. PEFT shows potential in adapting FMs for image recognition effectively in computer vision (CV) [26], [29], [35], while studies in remote sensing introduce PEFT paradigm to adapt FMs for dense prediction tasks which are more prominent in the geoscience community [23], [24]. AiRs [23] proposes the first PEFT framework for remote sensing image interpretation, which showcases the capability to surpass the performance of full fine-tuning and other strong PEFT methods with less than 5% of trainable parameters.

While most PEFT methods in remote sensing and CV do reduce the trainable parameters and maintain comparable performance, they are not that training-efficient when compared to full fine-tuning [23], [30], [31], [32], [33]. While resource-rich researchers can afford the tuning cost of FMs, the training efficiency (especially the tuning memory footprint) is still a bottleneck for most resource-limited researchers. The training efficiency of PEFT, especially in terms of training memory footprint, has not been explored extensively in remote sensing.

Received 21 March 2024; revised 15 August 2024 and 9 October 2024; accepted 23 October 2024. Date of publication 1 November 2024; date of current version 14 November 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2021YFB3900504; and in part by the National Natural Science Foundation of China under Grant 62201550, Grant 62331027, and Grant 62171436. *(Corresponding authors: Wanxuan Lu; Hongfeng Yu.)*

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Digital Object Identifier 10.1109/TGRS.2024.3488731

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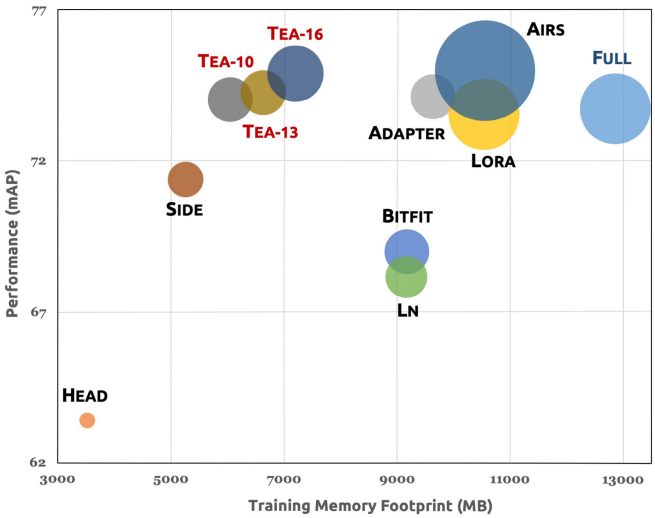


Fig. 1. Comparisons of training efficiency between TEA and other tuning methods. In DIOR, TEA can save about 50% of the training memory footprint and 15% of the training time compared with full fine-tuning, while achieving superior performance. We can say that TEA achieves an impressive tradeoff between training efficiency and performance. The area of the circle represents the training time cost.

Most existing PEFT arts still fully execute the forward and backward processes of FMs [25], [26], [29], [30], [31], [36]. Thus, the intermediate activations stored for backpropagation and the computations of backward are not omitted. Enhancing the tuning efficiency is a key step in the widespread application of FMs. Side-tuning (ST) [37] first tries to reduce the training memory footprint by sidestepping the gradient flowthrough the pretrained convolution networks. However, methods like ST have typically found it challenging to achieve performance comparable to that of full fine-tuning [26]. As shown in our experiments, previous ST has also struggled to provide satisfactory performance as standard PEFT methods do on remote sensing datasets. Moreover, these "ST" methods have not sufficiently considered the strong demand for macroscopic scene perception in remote sensing scenarios. Perceiving the scene macroscopically is crucial for remote sensing tasks, as the remote sensing scene is usually grand and complex. The existing ST methods do not consider this demand, which may affect their performance in remote sensing tasks.

To alleviate the issues aforementioned, we propose TEA, a training-efficient adapting framework for remote sensing transfer learning. TEA stitches the extremely lightweight networks and the cumbersome FMs together to perform as well as FMs while maintaining the training and parameter efficiency of compressed networks. Fig. 1 shows TEA's superior tradeoff between training efficiency and performance over strong PEFT baselines. Specifically, TEA is composed of three main components and one macroscopic perception mechanism: 1) an off-the-shelf feature extractor. The feature extractor is generally a pretrained powerful visual FM (VFM) such as RingMo [17], SwinTransformer [38], and ConvNeXt [39]. It provides general representations learned in the pretraining stage. 2) A SIDE adapter network. We call it "SIDEAN." SIDEAN in TEA is generally a powerful lightweight network that can effectively and efficiently process remote sensing images and inject task-specific geographical information into the general representations provided by the frozen VFM. 3) Task-related components. The specific task-related components for each task, such as the task head and neck. They leverage the refined features from SIDEAN and perform the task. TEA constructs a training highway attached to the FM, which bypasses the backpropagation process of the cumbersome FM and makes TEA training-efficient. Moreover, TEA is parameter-efficient, as the trainable parameters of SIDEAN are generally much fewer than those of the FM. We recognize that remote sensing scenes are often vast and complex, and the identification and extraction of valuable geographic information often require a macroscopic understanding of the scene. We observe that the VFMs pretrained sufficiently are aware of distinguishable patterns, and features from higher layers are more fully equipped to understand remote sensing scenes. Thus, we propose a top-down guidance mechanism to make TEA aware of the macro remote sensing scenes by foreseeing the more informative features from higher VFM layers. Then, TEA attains the ability to perceive the remote sensing scenes macroscopically and elevates its performance to be competitive with SOTA tuning frameworks.

To validate the efficiency and generality of TEA, we conduct extensive experiments on ten commonly adopted datasets that cover four mainstream remote sensing downstream tasks. We can save over 50% of training memory usage compared with that of full fine-tuning and achieve superior performance in all test tasks. Moreover, TEA can save up to 15% of training time in most dense prediction tasks, and this percentage reaches up to 28% in the semantic segmentation task. When scaling up the feature extractor to Swin-L, TEA can save more than 60% of memory footprint and about 30% of training time while surpassing the performance of full fine-tuning. TEA is also very competitive in terms of performance, it surpasses full fine-tuning on all object detection datasets and the orientated object detection dataset. Moreover, We have tested TEA with a CNN-based VFM and achieved competitive performance, which demonstrates the generality of TEA.

Our contributions are summarized as follows.

1) We propose a TEA framework for remote sensing transfer learning. To the best of our knowledge, TEA is the first framework in remote sensing which improves both training and parameter efficiency of the fine-tuning process while achieving dependable performance across multiple mainstream remote sensing tasks.
2) Combining powerful FMs and efficient lightweight networks, TEA provides an impressive tradeoff between training efficiency and performance when applying FMs to remote sensing tasks. With the top-down guidance mechanism, TEA can perceive macroscopic scene information and achieve competitive performance to SOTA tuning methods.
3) TEA serves as a flexible and general framework for many FMs, lightweight networks (SIDEAN), and tasks. The only modification is to replace the corresponding components.

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We organize the remaining chapters as follows. In Section II, we list the pioneers related to our study. We describe our methods in detail in Section III. The experiments are reported in Section IV. Finally, we conclude our work in Section V.

II. Related Work

Here, we review the related work in three aspects: 1) FMs; 2) PEFT; and 3) training efficiency.

A. Foundation Models

Vision FMs (VFMs) have garnered significant interest in recent research endeavors. The objective of these models is to acquire comprehensive visual features through extensive pretraining, which enables them to perform admirably across a spectrum of downstream applications. Within the CV sphere, models like SwinTransformer [38] and ViT [40], which are pretrained with supervised learning on ImageNet [41], have demonstrated remarkable proficiency in diverse downstream tasks. Inspired by the success of self-supervised pretraining in natural language processing, MAE [42] has pioneered the introduction of this approach in CV, yielding commendable results in downstream tasks. Given the similarities between optical remote sensing imagery and natural images, it is observed that general VFMs extend their utility to remote sensing downstream tasks as well [1], [2], [3], [4], [23], [43], [44], [45], [46], [47], [48].

It is worth noting that to augment the efficacy of remote sensing tasks, research within the remote sensing community has begun to integrate the idiosyncrasies of remote sensing data with established pretraining methodologies from the field of general vision. And there are a plethora of commendable studies about FMs in remote sensing that have emerged in recent years [17], [20], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58], [59]. SkySense is a recent and comprehensive state-of-the-art (SoTA) remote sensing FM that employs multigranularity contrastive learning and geo-context prototype learning methods to model multimodal spatiotemporal remote sensing data and have achieved exceptional performance in a multitude of remote sensing downstream tasks [49]. SatLasPreTrain has established a large-scale multilabel spatiotemporal remote sensing pretraining dataset based on Sentinel-2 and NAIP data sources, demonstrating that pretraining models on remote sensing data significantly aids in the performance of remote sensing tasks [55]. Based on the MAE pretraining paradigm, SatMAE pretrains on temporal and multispectral remote sensing images, enhancing the performance of downstream tasks such as scene classification and building segmentation [56]. RingMo applies the MIM pretraining paradigm and is pretrained on a large-scale optical remote sensing image dataset [17]. On the other hand, MTP proposes a novel multitask pretraining (MTP) paradigm to mitigate the discrepancy between pretraining tasks and downstream tasks and constructs a dataset (SAMRS) for pretraining [51]. Collectively, these arts have meaningfully propelled the progress and application of FMs within the remote sensing domain.

The advancements in both general and remote sensing FMs have prompted us to realize that how to efficiently apply these powerful models to remote sensing downstream tasks is a topic well worth investigating. In our research, we have selected SwinTransformer, RingMo, and MTP for exploration.

B. Parameter-Efficient Fine-Tuning

The concept of PEFT has garnered significant attention within the FM era due to its dual advantages of reducing trainable parameters' volume while maintaining comparable performance.

Adapter [30] is the first work to propose the concept of PEFT. It introduces lightweight adaptation blocks to the pretrained model and updates only the parameters in these blocks during fine-tuning. LoRA [31] proposes to reparameterize some layers in the pretrained model with low-rank matrices during fine-tuning and merge the extra branches into the original backbone in inference. It achieves a better tradeoff between the parameter and inference efficiency. Compactor [60] focuses on compressing the volume of trainable parameters further by replacing the up- and down-projection weights with low-rank parameterized hypercomplex multiplication (LPHM) layers. BitFit [32] and LN-tuning [33] do not modify or reparameterize the pretrained model but only reassign selected parameters during fine-tuning. Specifically, BitFit updates "bias" parameters only, while LN-tuning adjusts the "normalization" layers' parameters. PEFT has also been widely studied in the fields of image recognition [26], [27], [28], [35] and video understanding [29]. VPT [26] provides the first systematic exploration of the potential of vision PEFT. It allocates trainable prompt tokens to original image tokens and just updates these tokens during training. AiRs [23] proposes the first PEFT framework for tuning the remote sensing downstream tasks and showcases its capability to surpass the performance of full fine-tuning.

PEFT does exhibit extremely high parameter efficiency and satisfactory performance, but it remains suboptimal in terms of training efficiency. Generally, PEFT methods do not save training memory footprint significantly.

C. Training Efficiency

Training efficiency can be measured by the memory footprint and training time. Head-tuning [61] is extremely efficient in terms of training memory usage and time cost, but it does not provide satisfactory performance. Gradient checkpoint [62] is proposed to reduce the memory footprint at the cost of recomputing the intermediate activations during back-propagation. The gradient checkpoint trick is not parameter-efficient, as it only saves the training memory footprint but not the trainable parameters' volume.

Turning to lightweight models [63], [64], [65], [66], [67] is another way to improve training efficiency. By employing the lightweight models, the training cost will be reduced significantly, as the much fewer parameters and lower dimensions of the efficient models. However, the performance of lightweight models is usually inferior to the large pretrained models. To enhance the performance of lightweight models, some work applies the knowledge distillation framework [3], [4], [68], [69] to align the output distribution of the small models to the

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teacher models. However, an extra training process is required to obtain the teacher model.

To pursue a better tradeoff among parameter volume, training efficiency, and performance, ST [37] conducts the first exploration to avoid the high memory footprint by sidestepping the gradient flowthrough the pretrained model. Thus, a large portion of the training memory requirements due to intermediate activations and parameter gradients are avoided. LST [70] shares a similar idea and shows reliable results on NLP tasks. LST is designed for encoder–decoder language models, which is not suitable for vision backbone. In addition, some noteworthy studies have explored the potential of ST-like strategies from the perspective of optimizing specific tasks. For example, SAN [71] applies this strategy to open-vocabulary semantic segmentation and achieves superior performance, while BAN [72] uses this paradigm to leverage pretrained model to enhance the performance of change detection baselines.

The aforementioned ST-like PEFT methods are training-efficient but mostly perform inferior to the full fine-tuning. Different from these methods, TEA stitches the FMs and the lightweight models together and treats the lightweight models as powerful feature refiners. Thus, TEA can achieve competitive performance to the full fine-tuning while maintaining similar training efficiency to the lightweight models.

III. Method

First, we will analyze the potential cost during fine-tuning and discuss the strategy to relieve this issue. Then, we introduce the overall framework of our TEA. And we will elaborate on the details of the side adapter (SIDEAN). Finally, we will discuss the top-down guidance mechanism of TEA.

A. Preliminaries

We review the forward and backpropagation process of full fine-tuning and adapter-tuning. Then, we analyze the potential memory usage during the tuning stage. We show the overview of the forward and backpropagation process of different tuning methods in Fig. 2.

1) Backpropagation of Full Fine-Tuning: We take a standard vision transformer to block [40] to explain the forward and backpropagation process here. Given the ℓth transformer block that consists of an multihead self-attention (MSA) layer $\mathcal{F}^{\ell}_{m\mathfrak{sa}}(\cdot, \theta^{\ell}_{m\mathfrak{sa}})$ and an MLP layer $\mathcal{F}^{\ell}_{m\ell p}(\cdot, \theta^{\ell}_{m\ell p})$ and the input $x_{\ell-1}$, we have the forward process as follows:

$$\begin{aligned} x_{\ell'} &= \mathcal{F}^{\ell}_{m\mathfrak{sa}}\left(x_{\ell-1}, \theta^{\ell}_{m\mathfrak{sa}}\right) \\ x_{\ell} &= \mathcal{F}^{\ell}_{m\ell p}\left(x_{\ell'}, \theta^{\ell}_{m\ell p}\right) \end{aligned} \tag{1}$$

where $\theta^{\ell}_{m\mathfrak{sa}}$ and $\theta^{\ell}_{m\ell p}$ are the trainable parameters of corresponding layers, and $x_{\ell'}$ is the intermediate activation inside the ℓth vision transformer block. We omit the specific dimensions of the tensors and weights. To update the $\theta^{\ell}_{m\mathfrak{sa}}$ in $\mathcal{F}^{\ell}_{m\mathfrak{sa}}(\cdot, \theta^{\ell}_{m\mathfrak{sa}})$, we have the following gradient chain:

$$\frac{\partial \mathcal{L}}{\partial \theta^{\ell}_{m\mathfrak{sa}}} = \frac{\partial \mathcal{L}}{\partial x_{\ell}} \cdot \frac{\partial \mathcal{F}^{\ell}_{m\ell p}}{\partial x_{\ell'}} \cdot \frac{\partial \mathcal{F}^{\ell}_{m\mathfrak{sa}}}{\partial \theta^{\ell}_{m\mathfrak{sa}}} \tag{2}$$

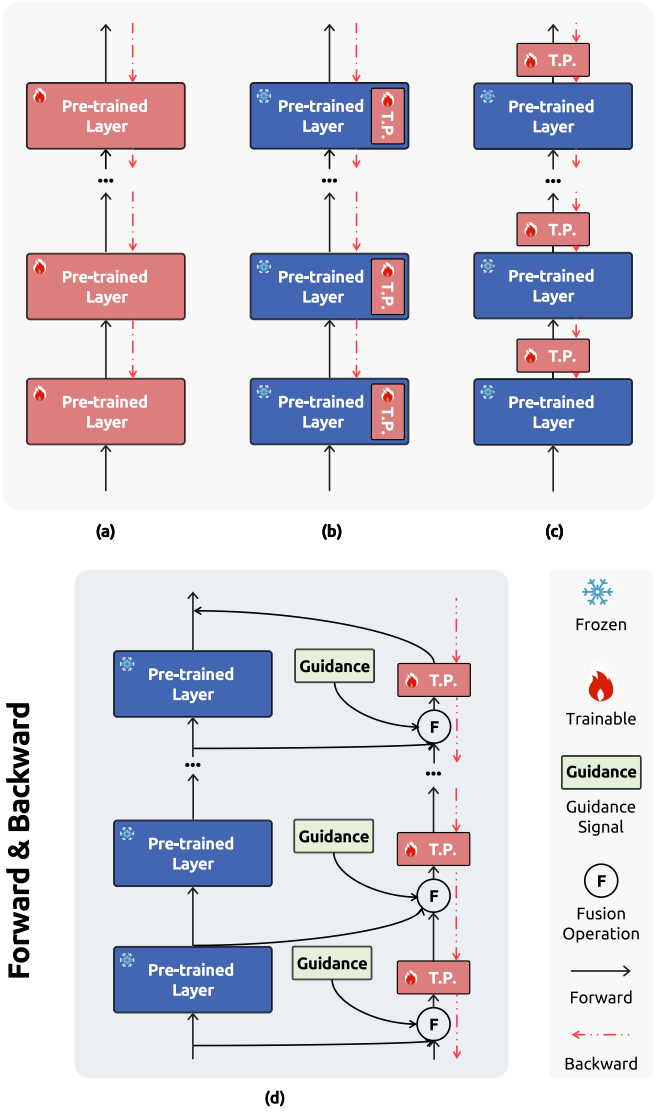


Fig. 2. Overview of the forward and backward processes of different tuning methods. (a) Full fine-tuning. The forward and backward processes are executed on all pretrained layers. (b) Selection-based PEFT methods. The training process is similar to (a) while only updating the selected parameters (e.g., BitFit and LN). (c) Addition-based PEFT methods. The adapter is the most representative method of this category. These methods do not change the training paths but only update the additional parameters. (d) TEA. Our TEA pipeline only executes the forward process and sidesteps the backpropagation process of the off-the-shelf VFMs. "T.P." means the trainable parameters.

where $\mathcal{L}$ denotes the task loss. When expanding the $((\partial \mathcal{L})/(\partial x_{\ell}))$ to mth layer ($m > \ell$), we have

$$\frac{\partial \mathcal{L}}{\partial \theta^{\ell}_{m\mathfrak{sa}}} = \frac{\partial \mathcal{L}}{\partial x_{m}} \cdot \frac{\partial \mathcal{F}^{m}_{m\ell p}}{\partial x_{m'}} \cdot \frac{\partial \mathcal{F}^{m}_{m\mathfrak{sa}}}{\partial x_{m-1}} \cdots \frac{\partial \mathcal{F}^{\ell}_{m\ell p}}{\partial x_{\ell'}} \cdot \frac{\partial \mathcal{F}^{\ell}_{m\mathfrak{sa}}}{\partial \theta^{\ell}_{m\mathfrak{sa}}}. \tag{3}$$

In most deep learning frameworks, to get the gradient of $\theta^{\ell}_{m\mathfrak{sa}}$, all relevant intermediate activations in the backpropagation chain shown in (3) have to be preserved. What's more, the gradient of $\theta^{\ell}_{m\mathfrak{sa}}$ and the trainable parameters in the chain are also required to be stored.

2) Backpropagation of Adapter-Tuning: We take adapter [30] as an example to explain the training process, and then extend it to other popular PEFT methods. Adapter inserts extra trainable blocks into the pretrained MSA $\mathcal{F}^{\ell}_{m\mathfrak{sa}}(\cdot, \theta^{\ell}_{m\mathfrak{sa},\mathfrak{ada}p})$ and MLP $\mathcal{F}^{\ell}_{m\ell p}(\cdot, \theta^{\ell}_{m\ell p,\mathfrak{ada}p})$. Generally,

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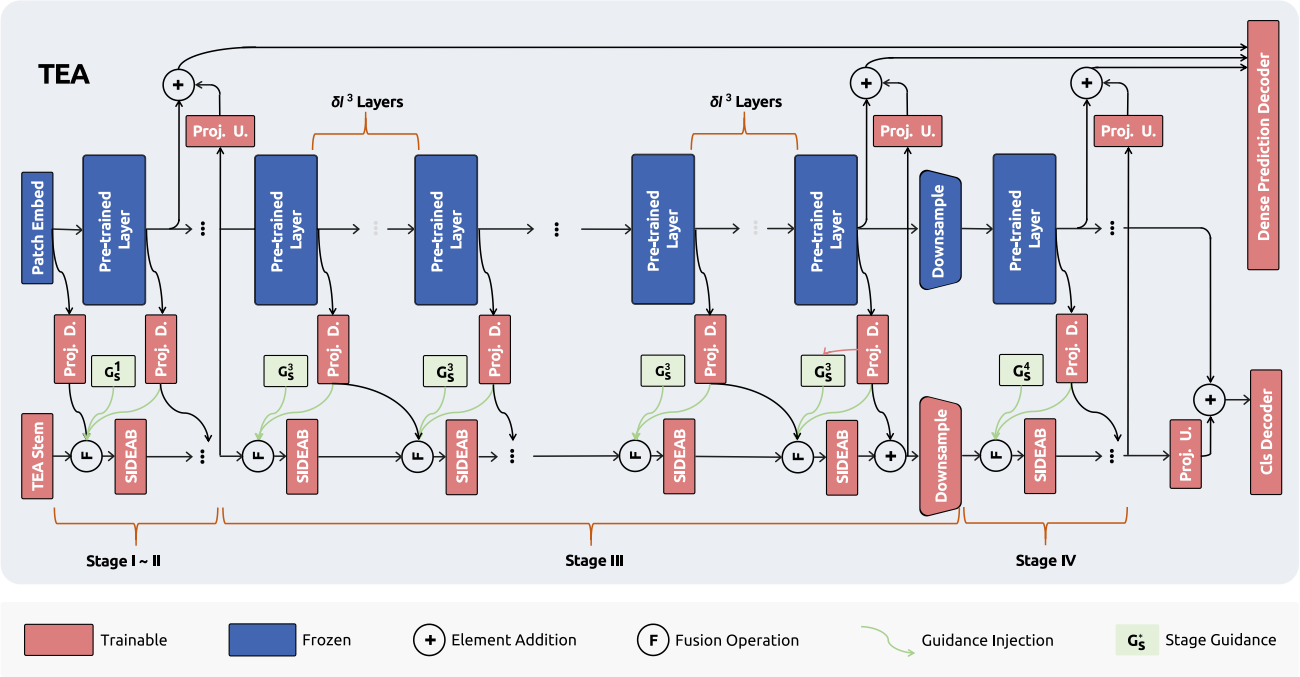


Fig. 3. TEA framework. (a) Pipeline. Inputs are fed into the pretrained patch embed layer and TEA stem, respectively, and then passed to the pretrained backbone and SIDEAN. VFM features and guidance signals are injected into SIDE adapter blocks ("SIDEABs" in the figure) by fusion operations. The task-required features from both paths are added before being fed into the decoder. (b) Trainable parameters. All TEA components in the figure above are set to be trainable. TEA stems, SIDEABs, downsample layers in the TEA path, and task-related decoders are trainable. (c) Notes. We omit the "normalization" layers and the scaling coefficients in "element addition" and "fusion" operations for clarity of expression. In fact, we add a layer normalization (LN) layer for each injection of the guidance signal, and a scaling coefficient is attached to each VFM feature and guidance signal. (d) Symbol legends. "Proj. D." means the down projection, while "Proj. U." refers to the up projection. "SIDEAB" here means the blocks of SIDEAN for refining the features.

$|\theta^{\ell}_{msa,ada}|$ and $|\theta^{\ell}_{mlp,ada}|$ are much smaller than $|\theta^{\ell}_{msa}|$ and $|\theta^{\ell}_{mlp}|$. The forward process is given as follows:

$$x_{\ell'} = \mathcal{F}^{\ell}_{msa}\left(x_{\ell-1}, \theta^{\ell}_{msa,ada}\right)$$
$$x_{\ell} = \mathcal{F}^{\ell}_{mlp}\left(x_{\ell'}, \theta^{\ell}_{mlp,ada}\right). \tag{4}$$

We omit the specific operations inside the MSA and MLP layers, as the adapter-tuning does not change the main operations and the output format of the original blocks. The main difference between (3) and (4) is the parameterization of trainable parameters. In (3), all pretrained parameters are set to trainable, while in (4), only the lightweight adapters are updated. Following (3), we have the backpropagation of adapter-tuning as

$$\frac{\partial \mathcal{L}}{\partial \theta^{\ell}_{msa,ada}} = \frac{\partial \mathcal{L}}{\partial x_m} \cdot \frac{\partial \mathcal{F}^{m}_{mlp}}{\partial x_{m'}} \cdot \frac{\partial \mathcal{F}^{m}_{msa}}{\partial x_{m-1}} \cdots \frac{\partial \mathcal{F}^{\ell}_{mlp}}{\partial x_{\ell'}} \cdot \frac{\partial \mathcal{F}^{\ell}_{msa}}{\partial \theta^{\ell}_{msa,ada}}. \tag{5}$$

Comparing (4) [see Fig. 2(c)] and (3) [see Fig. 2(a)], we can see that the backpropagation of adapter-tuning is almost the same as the full fine-tuning. So the memory usage for intermediate activations cannot be reduced significantly. Most memory savings of adapter-tuning come from the compressed trainable parameter volume, as the gradients for most pretrained parameters are avoided.

The analysis above can be naturally extended to LoRA, BitFit, and other SOTA PEFT methods [23], [27], [28], [29], [31], [32], [33], [36]. The $\theta^{\ell}_{msa,ada}$ of LoRA are the parameters reparameterizing the frozen linear projection layers. For BitFit, $\theta^{\ell}_{msa,ada}$ and $\theta^{\ell}_{mlp,ada}$ are "bias" items in the pretrained blocks. In general, the macro memory usage is similar, as long as the pretrained VFM layers are in the backpropagation chain.

B. Training-Efficient Adapting

1) Overall Pipeline: TEA framework targets to improve the training efficiency of the fine-tuning while maintaining the parameter efficiency and competitive performance. TEA framework is composed of three parts: 1) an off-the-shelf feature extractor; 2) a SIDEAN; and 3) task-related components. We show the overall pipeline of TEA in Fig. 3. Input images are first processed by the feature extractor (the "Pretrained Layer"s in Fig. 3) to obtain general visual representations. Then, the SIDEAN ("SIDEAN" in the figure above) attached to the extractor refines the features for target downstream tasks. Finally, the task-related components (the task head, and neck) perform the final prediction based on the refined features.

In TEA, the feature extractor is generally a VFM, which "is any model that is trained on broad data (generally using self-supervision at scale) that can be adapted (e.g., fine-tuned) to a wide range of downstream tasks." [16]. Our experiments in Section IV support the generality of TEA for VFMs. During tuning, the feature extractor is locked and only executes the forward process to output general vision representations. It is worth noting that the backpropagation is not performed through the feature extractor, and thus the extractor-related training memory footprint is significantly reduced.

As the gradient flowthrough the feature extractor is blocked, a large portion of the training cost is avoided. Take head-tuning [61] as the most training-efficient example, the training

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memory footprint is only 30% while the training overhead is just 50% of the full fine-tuning (as shown in Table III). However, the performance drops a lot as the capability of the feature extractor is constrained. To compensate for the loss caused by the skipped gradient flow, we propose to attach a SIDEAN to the locked backbone and make the distribution of the extractor output learnable. The SIDEAN in TEA is a lightweight structure that adapts the general vision features to the target downstream tasks. Indeed, we choose a well-designed lightweight net (GhostnetV2 [63]) as the SIDEAN for TEA. The off-the-shelf feature extractor with SIDEAN can be viewed as the trainable backbone in a task framework. We only update the SIDEAN to align the distribution of the backbone output with the intrinsic distribution of the downstream task. Training a SIDEAN is usually efficient and resource-saving, and the lightweight structure is always with a few parameters. Thus, TEA is both training- and parameter-efficient.

TEA is a general tuning framework that can be applied to various downstream tasks. The task-related components can be selected flexibly. We adopt TEA for four commonly studied remote sensing tasks: object detection, semantic segmentation, scene classification, and orientated object detection. TEA showcases extremely high training efficiency and competitive performance across all these tasks.

2) Backpropagation of TEA: We formulate the training process of TEA similarly as Section III-A, and omit the feature projection between the SIDEAN and the feature extractor. The pretrained MSA and MLP are given as $\mathcal{F}_{msa}(\cdot)$ and $\mathcal{F}_{mlp}(\cdot)$, respectively. There are no θ in these formulas, as we set pretrained MSAs and MLPs to be frozen in TEA. The nth SIDEAN block is formulated as $\mathcal{F}^n_{sidean}(\cdot, \theta^n_{sidean})$ and n is not always equal to the ℓ in (1). Then, the forward process of a single SIDEAN block is

$$\begin{aligned} x_{\ell-\delta\ell^s} &= \mathcal{F}^{\ell-\delta\ell^s}_{mlp}\left(\mathcal{F}^{\ell-\delta\ell^s}_{msa}(x_{\ell-\delta\ell^s-1})\right) \\ x'_{n'} &= \mathcal{F}^n_f\left(x'_{n-1}, x_{\ell-\delta\ell^s}, x_{guid}\right) \\ x'_n &= \mathcal{F}^n_{sidean}\left(x'_{n'}, \theta^n_{sidean}\right) \end{aligned} \tag{6}$$

where $x_{\ell-\delta\ell^s}$ refers to the output of the $\ell-\delta\ell^s$ th pretrained layer. $\delta\ell^s$ means the $\delta\ell$ in stage s. We use x'_n to refer to the output of the n th SIDEAN block (SIDEAB). $x'_{n'}$ is a temporary variable to illustrate the output of the "fusion operation." TEA modifies the output of a pretrained layer by a side block, which is not inserted into the original layer. The input for the SIDEAN block is first fused by $\mathcal{F}_f(\cdot)$, e.g., the fusion operation in Figs. 2 and 3, and we will explain this in Section III-D. We omit the updating parameters in $\mathcal{F}_f(\cdot)$ here, as it does not affect the main gradient chain and also for the sake of clarity in expression. We have the backpropagation of TEA as

$$\frac{\partial \mathcal{L}}{\partial \theta^n_{sidean}} = \frac{\partial \mathcal{L}}{\partial x'_m} \cdot \frac{\partial \mathcal{F}^m_{sidean}}{\partial x'_{m-1}} \cdots \frac{\partial \mathcal{F}^n_{sidean}}{\partial \theta^n_{sidean}}. \tag{7}$$

According to the formula above and Fig. 2(d), we can see that TEA simplifies the backpropagation chain. Then, the memory occupied by redundant activations is avoided, and the time cost of the backward process is reduced. Also, the dimension of intermediate activations in the SIDEAN is much smaller than that of VFMs. Moreover, the number of trainable parameters is very small, which means the memory usage related to parameter volume can also be reduced. We can say that TEA is training- and parameter-efficient.

C. SIDEAN

SIDEAN is the core tuner in TEA. With SIDEAN, we construct a training highway to bypass the burden of training large VFMs and maintain the capability of the powerful backbone. In this section, we will introduce the design of the SIDEAN in detail.

As we analyze in Section III-B2, the limited memory saving of most PEFT baselines is attributed to the unblocked gradient flowthrough the large VFMs [23], [30], [31], [32], [33]. To overcome this pain point, we propose to attach a SIDEAN to the off-the-shelf extractor to unlock the capability of VFMs while bypassing the burden of training the large VFM. Specifically, we pass the raw remote sensing images to the SIDEAN and inject intermediate features of the frozen VFM into the SIDEAN as shown in Fig. 3. By combining the frozen extractor and SIDEAN closely, the SIDEAN can understand the remote sensing images from bottom to top and adapt the features step-by-step according to the feedback of task loss, which is crucial for unlocking the capability of the frozen VFM.

To combine the VFM with the SIDEAN, we simply adopt the standard linear layers to map the high-dimensional features to the SIDEAN feature space. As shown in Fig. 3. Then, the task-orientated features refined by the SIDEAN will be mapped back to the VFM feature space and fused with the original representations. In addition, we place normalization layers (LN) before the "Proj. D."s and after the output fusion to stabilize the training process. It's noteworthy that we only map the specific features asked by the downstream framework back to the high-dimensional space, significantly reducing the memory requirements for retaining backpropagation-related intermediate activations. To shorten the forward path and pursue better training efficiency, we adopt a strategy similar to LayerDrop [73] in TEA, and we do not pass the features of all pretrained layers to the SIDEAN. Specifically, We set $\delta\ell^s$ to denote the number of pretrained layer intervals between two neighboring trainable adapting blocks in stage s and determine the selection of features from the VFM. By default, we configure $\delta\ell^s$ according to the pretrained model and the specific setup of the SIDEAN we employ, setting $\delta\ell^s$ individually for each stage s of the FM. Taking stage 3 of Swin-B and GhostnetV2 for example, with Swin-B comprising 18 layers and GhostnetV2 consisting of 5 layers, we calculate $\delta\ell^3$ to be $18/5 = 3.6$, adopting 4 as our rounded value. Consequently, the Swin-B layers selected are {5, 9, 13, 17, 21} for stage 3, indexed according to Swin-B's original layer indices. As feature selection plays a vital role, we have studied the different selection strategies. Finally, we find that uniformly selecting layers according to $\delta\ell^s$ within each stage based on VFM's inherent stage division works best. We adopt this rule in all of our experiments. The ablation study is given in Section IV-F.

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In theory, TEA does not impose specific limitations on the structure of SIDEAN. The sole requirement of SIDEAN is its capability to effectively handle remote sensing images and inject extracted remote sensing-related information into the generic features provided by VFMs. We tried two settings for SIDEAN, e.g., stacked residual blocks and structured lightweight networks. As reported by the results in Table IX, the thoughtfully designed lightweight networks are the optimal choice for SIDEAN. Among the lightweight networks we tested, GhostnetV2 [63] performs best and shows satisfactory training efficiency. Thus, we adopt GhostnetV2 as the default SIDEAN in all of our experiments. The blocks and downsample layers in GhostNetV2 correspond to the SIDEAN blocks and downsample blocks in Fig. 3. It is worth noting that, for "TEA Stem" in Fig. 3, we adopt the first several layers in GhostnetV2 which downsample the input before the main GhostnetV2 blocks.

D. Top-Down Guidance

Just employing the SIDEAN to tune the extracted features from the VFM in a naive way does not show powerful enough performance compared to SOTA PEFT methods, such as AiRs and adapter, as shown in Table X. We attribute the relatively poor performance to the weak perception of the grand remote sensing scenes. In remote sensing tasks, a key point is to extract valuable information from the massive background, which relies on the large receptive field and the high-level understanding of the images. The small parameter volume of the lightweight SIDEAN and the efficient but relatively limited LayerDrop mechanism constrain the capability of the SIDEAN to understand the grand scene of remote sensing images. Thus the SIDEAN cannot extract the target-related information from the massive background features effectively. To overcome this issue and enhance the capability of TEA, we propose a top-down guidance mechanism to guide the training of SIDEAN with high-level features from pretrained VFMs. We visualize the feature maps of RingMo in Fig. 4 and observe that the pretrained VFMs have already captured some distinguishable patterns of the scenes and the semantic concepts of these patterns become more evident closer to the top layers. Thus, we pass the high-level features of pretrained VFMs as guidance signals to the SIDEAN to enhance the macro perception ability of the lightweight SIDEAN

$$x_\ell / x'_n = \mathcal{F}^n_{adap}\left(x_{\ell-\delta\ell^\jmath}, x_{aux}, \theta^n_{adap}\right). \tag{8}$$

We formulate the abstract adapting process of a pretrained feature as (8). x_ℓ denotes the feature to be refined. x_{aux} represents the auxiliary input, which is generally "None" for standard PEFT methods and full fine-tuning, and is set as features from other SIDEAN modules or extractor layers for TEA. $\mathcal{F}^n_{adap}$ is the adapting or tuning module, and n is the index of the adapting module. We correspond the nth adapting block with the ℓth pretrained layer. Specifically, for TEA, the data flow of the VFM and the SIDEAN is separate, hence we apply a distinct notation x'_n to represent the data flow of the SIDEAN. In contrast, for typical PEFT methods like adapter, the adapting block is usually embedded within the pretrained

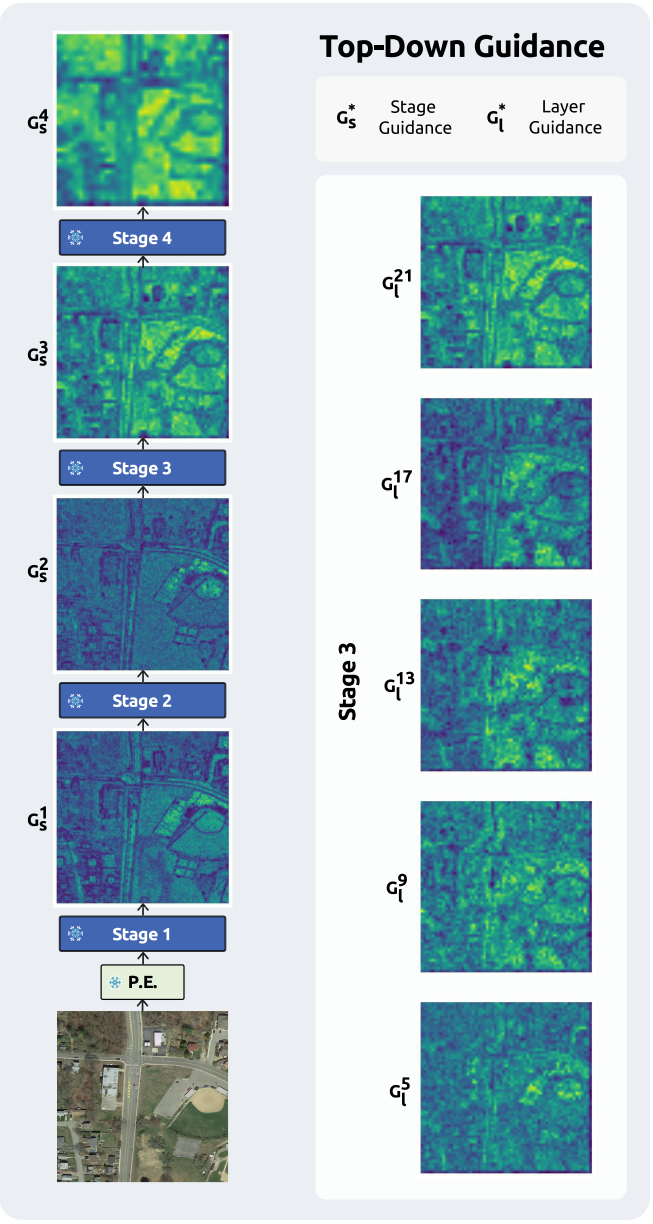


Fig. 4. Visualization of top-down guidance signals. We visualize the feature maps of pretrained RingMo. For clarity, we only show the feature maps of each stage (e.g., the stage guidance) and the feature maps in stage3 (e.g., the layer guidance). According to the visualization results, we can see that the pretrained model has already captured some distinguishable patterns of the scenes and the semantic information of these patterns becomes more evident closer to the top layers.

layers. Therefore, we consider the adapting block and its embedded pretrained layer as a whole, and use x_ℓ to denote the overall data flow. We treat the trainable parameters with the corresponding frozen pretrained layer of standard PEFT methods (e.g., adapter, LoRA, and BitFit) as a whole and denote it as the $\mathcal{F}^n_{adap}$ here. We set $\delta\ell^\jmath$ to denote the number of pretrained layer intervals between two neighboring trainable adapting blocks in stage $\jmath$. In the context of prevalent PEFT approaches such as Adapter and LoRA, $\delta^\jmath_\ell$ is conventionally set to 1, with the quantity of adapting blocks corresponding to the layer count of the pretrained model, denoted as $n = \ell'$. These tuning methods operate exclusively with the output from the preceding pretrained layer with the adapting block, as shown

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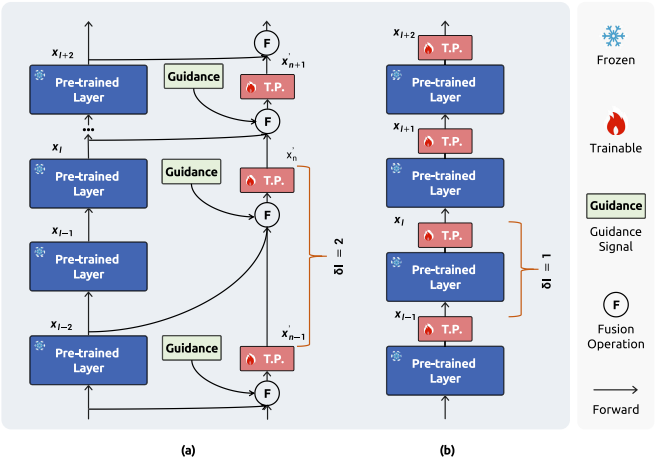


Fig. 5. $\delta\ell$ settings of TEA and adapter. $\delta\ell$ denotes the number of pretrained layer intervals between two neighboring trainable adapter blocks. In TEA, $\delta\ell$ is generally larger than 1, while in adapter, $\delta\ell$ is usually equal to 1. We take the adapter as an example here, and the settings for LoRA, BitFit, and LN are almost the same as for the adapter. We set $\delta\ell$ to be 2 in (a) just for illustrative purposes, the actual value needs to be set according to the specific model. "T.P" means trainable parameters, SIDEAB for TEA, and adapter blocks for adapter. (a) $\delta\ell$ setting in TEA. (b) $\delta\ell$ setting in adapter.

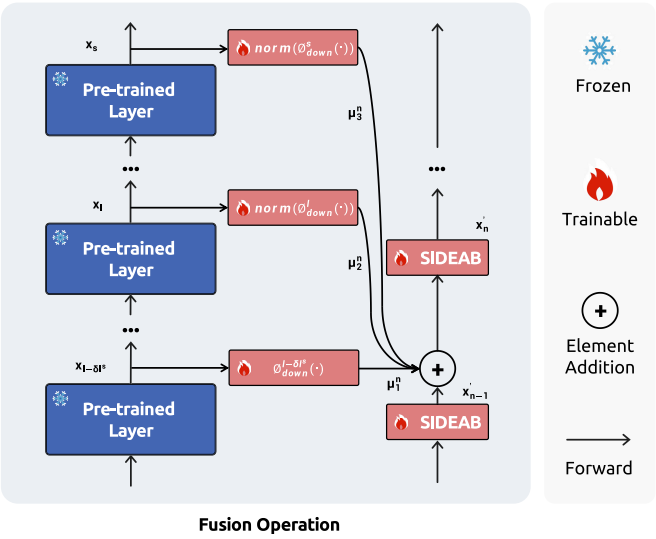


Fig. 6. Schema of fusion operation. This figure shows the details of the fusion operation in TEA, which is formulated as the second equation in (9).

in Fig. 5(b), which means the forward and backward process is performed layer-by-layer and the remote sensing-related information can be injected step-by-step. Thus, these tuning methods can align pretrained features with the requirements of remote sensing tasks more easily. Equation (8) is articulated as $x_\ell = \mathcal{F}^\ell_{adap}(x_{\ell-1}, \theta^\ell_{adap})$ in such cases. For TEA, as depicted in Fig. 5(a). $\delta\ell^s$ may not be confined to 1, thus necessitating a revised formulation of (8) to $x'_n = \mathcal{F}^n_{adap}(x_{\ell-\delta\ell^s}, x_{aux}, \theta^n_{adap})$. The adapting mechanism of TEA forces the adapter block in the SIDEAN to learn the amount of feature distribution modification which is achieved by several steps in other tuning methods.

As our analysis above indicates, the main challenge of TEA lies in its inability to gradually perceive and understand the vast remote sensing scenes, as SIDEAN is typically shallow and lightweight. To boost the adapting capability of the SIDEAN, we let the SIDEAN blocks foresee the macro scene of remote sensing images by passing high-level features as inputs to them. Notably, methods such as FPN [74] and UNet [75], along with their derivatives [43], [76], [77], have leveraged the fusion of features across various layers to enhance task performance. In addition, RevCol suggests that the transmission of high-level features from preceding subnetworks to the shallow layers of subsequent subnets can contribute to an iterative refinement of the features [78], [79]. Then, we callback (6) and give the full formulation of TEA as

$$\begin{aligned} x_{\ell-\delta\ell^s} &= \mathcal{F}^{\ell-\delta\ell^s}_{merge}\left(\mathcal{F}^{\ell-\delta\ell^s}_{merge}(x_{\ell-\delta\ell^s-1})\right) \\ x'_{n'} &= x'_{n-1} + \mu^n_1 \cdot \phi^{\ell-\delta\ell^s}_{down}(x_{\ell-\delta\ell^s}) + \mu^n_2 \cdot norm\left(\phi^\ell_{down}(x_\ell)\right) \\ &\quad + \mu^n_3 \cdot norm\left(\phi^s_{down}(x_s)\right) \\ x'_n &= \mathcal{F}^n_{sidean}\left(x'_{n'}, \theta^n_{sidean}\right) \end{aligned} \tag{9}$$

where we select the auxiliary input x_{guid} in (6) as the output of the ℓth pretrained layer x_ℓ (layer guidance), output of the current stage x_s (stage guidance). x'_n means the output of the n th SIDEAN block. $x'_{n'}$ is a temporary variable to illustrate the output of the "fusion operation." Other auxiliary input settings have been studied in Section IV-F. x_ℓ shows the processing mode of the current pretrained layer for the corresponding SIDEAN block. x_s provides the stage output as the macro guidance. $norm$ means the LN that we add after projection. ϕ^*_{down} refers to the projection layer of each feature, as shown in Fig. 6. For the process of TEA, we first adopt linear projection layers to map $x_{\ell-\delta\ell^s} \in \mathcal{R}^{D\times h\times w}$, $x_\ell \in \mathcal{R}^{D\times h\times w}$, and $x_s \in \mathcal{R}^{D\times h\times w}$ into the SIDEAN feature space. Then, we just add these inputs together with learnable coefficients and feed the result into an adaptation layer. In addition, if $x'_{n-1} \in \mathcal{R}^{d\times h\times w}$ then μ^n_1, μ^n_2, and μ^n_3 are set to be learnable weights corresponding to the nth SIDEAN block with the shape of $d \times 1 \times 1$. With top-down guidance, the performance of TEA is boosted as shown in Table X.

IV. Experiment

With the TEA framework, we unlock the representation capability of FMs while significantly controlling the training cost. To verify the superiority and effectiveness of TEA, we conduct extensive experiments using multiple FMs on four commonly studied remote sensing tasks: object detection, orientated object detection, semantic segmentation, and scene classification. We first introduce the experimental settings and then report the results of TEA on these tasks. Finally, we conduct ablation studies to analyze the effectiveness of TEA.

A. Experimental Settings

We give our experimental settings here. First, we introduce the datasets used in our experiments. Then, we describe the compared methods and the implementation details.

1) Datasets: Ten commonly used datasets have been adopted here to evaluate our TEA framework. For object detection, we select NWPU-VHR10, UCAS-AOD, and DIOR, the size of these three datasets gradually increases. For semantic segmentation, we choose the Vaihigen, LoveDA, and iSAID.

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TABLE I
DATASETS INFORMATION. "TRAIN SET" AND "TEST SET" ARE REPORTED AS OUT PRACTICAL SETTINGS. WE USE THE "PART1" TRAIN SET OF FAIR1M TO CONDUCT EXPERIMENTS

Dataset		Train Set	Test Set	Category
Cls.	UCM [80]	1050	1050	21
	AID [81]	2000	8000	30
	NWPU-RESISC45 [82]	6300	25200	45
Det.	NWPU-VHR10 [83]	640	160	10
	AOD [84]	1051	453	2
	DIOR [85]	11725	11738	20
Det.(OBB)	FAIR1M [86]	66564	212561	37
Seg.	Vaihingen[1]	344	398	6
	LoveDA [87]	4191	1796	7
	iSAID [88]	33978	11644	15

For scene classification, UCM, AID, and NWPU-RESISC45 are adopted. We conduct the OBB detection on FAIR1M, which is the currently largest and most fine-grained dataset. The details of these datasets are given in Table I.

In general, we follow the dataset settings of AiRs [23]. For FAIR1M, we train the model on the part1 train set and evaluate the trained model on the entire test set.

2) Compared Methods: We compare TEA with the traditional full fine-tuning and several SOTA PEFT methods. Specifically, we implement and report the results of the following methods.

a) Traditional tuning methods:

1) *Full Fine-Tuning (FULL):* All parameters of the pretrained model are set to trainable.
2) *Head-Tuning [61] (HEAD):* Only the task-related components are set to trainable, and the pretrained feature extractor is frozen entirely.

b) PEFT methods:

1) *ST [37] (SIDE):* A trainable side network is attached to the pretrained VFM, but ST only fuses the final output of the side network and the VFM, which is different from TEA.
2) *Adapter [30] (ADAPTER):* The pretrained model is inserted with extra trainable adapters, which are lightweight down- and up-projection residual layers.
3) *LoRA [31] (LORA):* Generally, pretrained "QKV" projection layers are reparameterized with trainable low-rank matrices.
4) *BitFit [32] (BITFIT):* The "bias" items in the pretrained blocks are set to be tunable during fine-tuning.
5) *LN-Tuning [33] (LN):* The "normalization" layers in the VFMs are set to be trainable, we call it LN-tuning as the "normalization" layers are usually "LayerNorm" in our settings.
6) *AiRs [23] (AIRS):* A recent PEFT method designed for remote sensing, which is the SOTA PEFT method for remote sensing tasks.

We adopt the training recipes of AiRs [23] for all methods except ST. For ST, we align the training recipe with TEA. Specifically, we adopt GhostnetV2 [63] as the side network for ST. For a fair comparison, we adopt the same downstream framework for each task. For object detection, we conduct all experiments with RetinaNet [89]. We employ Segformer [90] as the framework for semantic segmentation. The linear classification layer has been selected to conduct scene classification experiments. For OBB detection, we adopt the Rotated RetinaNet [89].

[1]https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-labelpotsdam.aspx

TABLE II
TRAINING RECIPES. "36" EPOCHS FOR DIOR AND UCAS-AOD AND "48" EPOCHS FOR NWPU-VHR10. "80K" ITERATIONS FOR LOVEDA AND ISAID AND "40K" ITERATIONS FOR VAIHIGEN. "800 × 800" FOR DIOR AND "896 × 896" FOR ISAID. AND THE REST OF THE DATASETS SHARE THE SAME SETTINGS

Task	Training Scheme	Epoch/Iter.	Batch Size	Input Size
Cls.	By Epoch	100	64	224 × 224
Det.	By Epoch	36 / 48	4	800 × 800 / 1000 × 600
Det. (OBB)	By Epoch	12	4	512 × 512
Seg.	By Iteration	80k/40k	8	896 × 896 / 512 × 512

3) Implementation Details: Our training settings are provided as follows.

a) Training architecture: We follow AiRs [23] and select SwinTransformer [38], RingMo [17], and MTP [51] as the feature extractor for main experiments. Specifically, Swin-B pretrained on ImageNet [41] is adopted for object detection, semantic segmentation, and scene classification. RingMo and MTP are also adopted to conduct these tasks. For OBB detection, we employ the pretrained RingMo [17] as the feature extractor. GhostnetV2 is selected as the default SIDEAN for all experiments. And we initialize the SIDEAN with the weights pretrained on ImageNet. We construct our downstream models based on the open-source code of OpenMMLab [91], [92], [93].

b) Training recipe: We select AdamW as the default optimizer for all tasks and tune the learning rate per dataset based on the default value in the OpenMMLab codebase. We tune other training settings per task and list the specific settings in Table II.

B. Results

We report the results of TEA on four remote sensing tasks in this section. We first introduce the results of TEA on object detection. Then, we report the results of TEA on OBB detection, semantic segmentation, and scene classification. For each task, we report the training memory footprint and time to show the training efficiency of TEA. The corresponding performance metrics are also reported to show the effectiveness of TEA. Reviewing the experimental results detailed in Tables III, V, and VI, we observe that TEA has demonstrated a strong performance trend regardless of the underlying pretrained model while operating with a notably lower training memory footprint, particularly in the domain of dense prediction tasks. Our exploration with ImageNet pretrained models suggests that TEA can compete favorably with full fine-tuning approaches, achieving results that are either on par or exceed those of full fine-tuning in some cases. For RingMo and MTP, TEA has shown promise on datasets where training data is somewhat constrained (those with fewer than 10 000 samples as noted in Table I), approaching or nearly matching the performance of full fine-tuning and surpassing

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TABLE III

RESULTS OF OBJECT DETECTION ON NWPU-VHR10, UCAS-AOD, AND DIOR. "↑" INDICATES THE HIGHER THE BETTER. "↓" INDICATES THE LOWER THE BETTER. WE TEST THE TRAINING MEMORY FOOTPRINT AND THE TRAINING TIME OF EACH METHOD ON THE SAME MACHINE AND REPORT THE TEST RESULTS ON DIOR. "IM1K-SP" MEANS THE EXTRACTOR IS SUPERVISEDLY PRETRAINED ON IMAGENET-1K AND "IM21K-SP" MEANS THE EXTRACTOR IS PRETRAINED ON IMAGENET-21K. FOR TEA AND PEFT METHODS, WE EMPLOY **"BOLD"** FONT TO HIGHLIGHT THE TOP METRIC, "UNDERLINE" FOR THE NEXT BEST METRIC, AND *"Italics"* FOR THE THIRD-RANKING METRIC. DIFFERENT COLORS ARE APPLIED TO HIGHLIGHT DIFFERENT EXTRACTORS, WITH THE MAPPING TABLE AS FOLLOWS: {IM1K-SP, IM21K-SP, RINGMO, MTP}

Methods	Pre-trained	Trainable Params		Training				Dataset (mAP) ↑		
				Memory		Time Cost		NWPU-VHR10	UCAS-AOD	DIOR
		# (M) ↓	% ↓	MB ↓	% ↓	h ↓	% ↓			
Traditional Tuning Methods										
FULL	IM1k-SP	86.75	100.00	12856	100.00	13.0	100.00	94.14	95.69	73.72
	IM21k-SP							94.88	95.94	76.40
	RingMo	86.75	100.00	12306	100.00	10.5	100.00	94.75	96.53	76.98
	MTP	95.53	100.00	16683	100.00	12.7	100.00	95.98	97.17	78.85
HEAD	IM1k-SP	0.00	0.00	3521	27.39	6.5	50.00	89.25	94.17	63.42
	IM21k-SP							91.89	95.24	67.76
	RingMo	0.00	0.00	3533	28.71	4.9	46.67	88.87	93.60	61.20
	MTP	0.00	0.00	3516	21.08	5.7	44.88	93.76	95.41	70.92
PEFT Methods										
SIDE	IM1k-SP	3.73	4.30	**5255**	**40.88**	10.0	76.92	93.86	95.14	71.38
	IM21k-SP							93.86	95.67	73.48
	RingMo	3.73	4.30	**5256**	**42.71**	7.9	75.24	92.16	94.29	71.06
	MTP	4.14	4.33	5370	32.19	8.5	66.93	95.21	96.10	71.21
ADAPTER	IM1k-SP	3.52	4.06	9626	74.88	11.0	84.62	95.10	96.10	74.12
	IM21k-SP							94.30	96.36	*76.40*
	RingMo	3.52	4.06	9156	74.40	8.9	84.76	94.51	96.20	76.88
	MTP	3.94	4.12	14221	85.24	10.9	85.83	96.90	96.53	77.02
LORA	IM1k-SP	3.50	4.04	10533	81.93	13.0	100.00	95.29	95.68	73.54
	IM21k-SP							95.72	96.43	75.75
	RingMo	3.50	4.04	8976	72.94	10.2	97.14	94.11	95.78	72.72
	MTP	3.95	4.13	14130	84.70	11.9	93.70	**97.67**	95.92	*75.75*
BITFIT	IM1k-SP	0.21	0.24	9176	71.38	11.0	84.62	88.58	95.79	68.99
	IM21k-SP							88.98	96.58	71.08
	RingMo	0.21	0.24	8678	70.52	8.9	84.76	93.88	95.74	70.37
	MTP	0.11	0.13	13910	83.38	10.8	85.04	95.49	95.98	75.61
LN	IM1k-SP	0.06	0.07	9166	71.30	10.7	82.31	93.56	95.83	68.15
	IM21k-SP							93.98	96.29	69.94
	RingMo	0.06	0.07	8671	70.46	8.5	80.95	*94.82*	95.74	68.24
	MTP	0.04	0.04	13914	83.40	10.5	82.68	95.04	95.77	75.16
AIRS	IM1k-SP	3.81	4.39	10545	82.02	14.5	111.54	*95.33*	**96.68**	**74.98**
	IM21k-SP							96.29	96.85	**76.99**
	RingMo	3.81	4.39	10081	81.92	11.5	109.52	**95.03**	*96.29*	**77.25**
	MTP	4.24	4.44	14758	88.46	12.3	96.85	97.10	*96.78*	**77.39**
Our Method										
TEA	IM1k-SP	4.71	5.43	6036	46.95	11.0	84.62	95.16	96.28	74.02
	IM21k-SP							95.11	*96.75*	75.73
	RingMo	4.71	5.43	6048	49.15	8.7	82.86	93.53	95.81	74.90
	MTP	4.98	5.21	7549	45.25	10.2	80.31	*96.57*	96.71	73.89
	IM1k-SP	7.41	8.54	*6620*	*51.49*	11.0	84.62	**96.12**	*96.45*	*74.25*
	IM21k-SP							*96.18*	**96.87**	76.20
	RingMo	7.41	8.54	*6645*	*54.00*	9.0	85.71	94.86	**96.92**	*75.36*
	MTP	7.82	8.19	*8149*	*48.85*	10.3	81.10	96.12	96.95	74.41
	IM1k-SP	10.73	12.37	7188	55.91	12.0	92.31	96.06	96.53	74.88
	IM21k-SP							**96.31**	96.74	76.53
	RingMo	10.73	12.37	7251	58.92	9.2	87.62	94.81	96.41	75.27
	MTP	11.32	11.85	8752	52.46	10.6	83.46	96.39	**97.13**	74.52

it on UCAS-AOD, NWPU-VHR10, UCM, AID, and NWPU-RESISC45.

1) Object Detection: We report the results of TEA on three commonly studied datasets in Table III. To explore

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TABLE IV

RESULTS OF ORIENTATED OBJECT DETECTION ON FAIR1M. "↑" INDICATES THE HIGHER THE BETTER. "↓" INDICATES THE LOWER THE BETTER. WE TEST THE TRAINING MEMORY FOOTPRINT AND THE TRAINING TIME OF EACH METHOD ON THE SAME MACHINE WITH A SINGLE 3080TI GPU AND REPORT THE TEST RESULTS ON FAIR1M WITH "BATCH SIZE" AS 2. "RINGMO" MEANS THE EXTRACTOR LOADS THE PRETRAINED WEIGHTS PROVIDED BY RINGMO [17]. FOR TEA AND PEFT METHODS, WE EMPLOY **"BOLD"** FONT TO HIGHLIGHT THE TOP METRIC, "UNDERLINE" FOR THE NEXT BEST METRIC, AND *"Italics"* FOR THE THIRD-RANKING METRIC

Methods	Pre-trained	Trainable Params		Training				Dataset (mAP) ↑
				Memory		Time Cost		FAIR1M
Swin-B		# (M) ↓	% ↓	MB ↓	% ↓	h ↓	% ↓	
Traditional Tuning Methods								
FULL	RingMo	86.75	100.00	9063	100.00	15.0	100.00	25.10
HEAD	RingMo	0.00	0.00	2193	24.20	7.2	48.00	19.11
PEFT Methods								
SIDE	RingMo	3.73	4.30	**3268**	**36.06**	11.2	74.67	22.83
ADAPTER	RingMo	3.52	4.06	6261	69.08	13.0	86.67	24.30
LORA	RingMo	3.50	4.04	6134	67.68	14.0	93.33	21.23
BITFIT	RingMo	0.21	0.24	5887	64.96	12.9	86.00	20.89
LN	RingMo	0.06	0.07	5887	64.96	12.4	82.67	21.09
AIRS	RingMo	3.81	4.39	7043	77.71	16.8	112.00	25.22
Our Method								
TEA	RingMo	4.71	5.43	3915	43.20	12.7	84.67	24.28
	RingMo	7.41	8.54	*4415*	*48.71*	13.8	92.00	*24.56*
	RingMo	10.73	12.37	4919	54.28	14.5	96.67	**25.28**

the scalability of TEA, we report three results corresponding to different scales of the SIDEAN, e.g., GhostNetV2 1.0×, GhostNetV2 1.3×, and GhostNetV2 1.6×. We primarily analyze the results of "IM1k-SP," and the conclusions remain the same for the results of "IM21k-SP." According to the results, we have the following observations.

a) TEA reaches superior performance with over 50% memory footprint savings: We first compare the performance of TEA with other tuning methods. As shown in Table III, TEA achieves very competitive performance to AIRS, which shows the best PEFT performance in several remote sensing tasks. Moreover, TEA surpasses FULL on all object detection datasets across different data volumes. For example, TEA achieves 76.53 mAP on DIOR, which is 0.13 higher than FULL and at the cost of only 55.91% memory usage. As for NWPU-VHR10 and UCAS-AOD, TEA (95.11 and 96.75) can even outperform FULL (94.88 and 95.94) by an obvious margin. From the perspective of memory footprint, HEAD provides the best memory efficiency but at the cost of the largest performance drop. SIDE also shows competitive memory efficiency, but the performance is not always that satisfactory when compared with other PEFT methods. The rest of the compared PEFT methods see similar memory footprint reductions, but not to a significant extent. As for the performance with "RingMo" and "MTP," we can see that TEA outperforms most PEFT methods with a competitive memory footprint. Among all the methods, TEA provides the best choice for tuning object detection models with a limited training memory budget. Specifically, it can achieve over 50% memory footprint savings while keeping competitive performance to the best tuning method.

b) TEA saves over 15% of the training time compared to FULL: We list the time cost in Table III and find that the PEFT methods are not always time-efficient. But all variants of TEA are all more time-efficient than FULL. When taking both the memory footprint and performance into account, we believe that TEA achieves a better tradeoff between training efficiency and performance.

c) TEA remains efficient in terms of parameter volume: We also report the parameter scale of each tuning method in Table III. We can see that TEA also shows competitive parameter efficiency and scales well with the parameter volume. And TEA demonstrates the lowest memory footprint with the comparable parameter volume. For example, ADAPTER, LORA, and AIRS occupy 1.59×, 1.75×, and 1.75× memory footprint of TEA at the similar level of trainable parameter volume (about 5.00%).

2) Orientated Object Detection: We follow the analysis of object detection. The results of FAIR1M are reported in Table IV. It is noteworthy that we adopt the pretrained weights provided by RingMo [17], which is pretrained on large-scale remote sensing datasets with the masked image modeling (MIM) proxy task.

a) TEA achieves the best performance on FAIR1M with the cost of only 54.28% memory footprint: FAIR1M is a challenging dataset for orientated object detection. All PEFT methods except AIRS and TEA perform inferior to the FULL. This phenomenon shows that challenging tasks call for stronger tuning methods in the low-parameter regime. Comparing AIRS and TEA, we can see that TEA surpasses AIRS with a much lower memory footprint and faster training speed. We can say that TEA provides a better option for tuning

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TABLE V

RESULTS OF SEMANTIC SEGMENTATION ON VAIHINGEN, LOVEDA, AND ISAID. "↑" INDICATES THE HIGHER THE BETTER. "↓" INDICATES THE LOWER THE BETTER. WE TEST THE TRAINING MEMORY FOOTPRINT AND THE TRAINING TIME OF EACH METHOD ON THE SAME MACHINE AND REPORT THE TEST RESULTS ON ISAID. "IM1K-SP" MEANS THE EXTRACTOR IS SUPERVISED PRETRAINED ON IMAGENET-1K AND "IM21K-SP" MEANS THE EXTRACTOR IS PRETRAINED ON IMAGENET-21K. FOR TEA AND PEFT METHODS, WE EMPLOY **"BOLD"** FONT TO HIGHLIGHT THE TOP METRIC, "UNDERLINE" FOR THE NEXT BEST METRIC, AND *"Italics"* FOR THE THIRD-RANKING METRIC. DIFFERENT COLORS ARE APPLIED TO HIGHLIGHT DIFFERENT EXTRACTORS, WITH THE MAPPING TABLE AS FOLLOWS: {IM1K-SP, IM21K-SP, RINGMO, MTP}

Methods	Pre-trained	Trainable Params		Training				Dataset (mIoU) ↑		
				Memory		Time Cost				
		# (M) ↓	% ↓	MB ↓	% ↓	h ↓	% ↓	Vaihingen	LoveDA	iSAID
Traditional Tuning Methods										
FULL	IM1k-SP	86.75	100.00	24254	100.00	20.9	100.00	81.64	51.87	67.47
	IM21k-SP							82.26	53.10	68.70
	RingMo	86.75	100.00	23961	100.00	17.1	100.00	82.78	51.31	67.04
	MTP	95.53	100.00	43532	100.00	11.8	100.00	81.79	52.12	67.06
HEAD	IM1k-SP	0.00	0.00	4702	19.39	12.3	58.85	78.02	47.77	58.32
	IM21k-SP							77.66	48.69	60.82
	RingMo	0.00	0.00	7621	31.81	8.0	46.78	76.13	44.94	40.82
	MTP	0.00	0.00	6920	15.90	5.0	42.37	73.65	46.99	49.89
PEFT Methods										
SIDE	IM1k-SP	3.73	4.30	**9357**	**38.58**	15.9	76.08	80.71	51.02	64.08
	IM21k-SP							80.96	51.45	65.65
	RingMo	3.73	4.30	**9350**	**39.02**	11.3	66.08	80.00	50.50	62.54
	MTP	4.14	4.33	**13048**	**29.97**	8.5	72.03	79.80	50.93	63.81
ADAPTER	IM1k-SP	3.52	4.06	18338	75.61	17.3	82.78	81.41	52.22	65.70
	IM21k-SP							82.19	53.13	*67.65*
	RingMo	3.52	4.06	17994	75.10	14.1	82.46	81.67	51.09	61.28
	MTP	3.94	4.12	38124	87.58	10.3	87.29	79.74	51.18	61.81
LORA	IM1k-SP	3.50	4.04	17852	73.60	24.2	115.79	81.49	51.93	65.44
	IM21k-SP							81.86	*53.07*	66.66
	RingMo	3.50	4.04	17502	73.04	16.3	95.32	80.60	48.77	54.64
	MTP	3.95	4.13	37922	87.11	11.0	93.22	77.34	49.07	58.36
BITFIT	IM1k-SP	0.21	0.24	17404	71.76	17.5	83.73	79.39	49.93	62.77
	IM21k-SP							80.52	51.20	64.36
	RingMo	0.21	0.24	17072	71.25	14.3	83.63	80.33	49.37	53.62
	MTP	0.11	0.13	37440	86.01	10.2	86.44	77.42	49.44	58.57
LN	IM1k-SP	0.06	0.07	17402	71.75	17.0	81.34	79.39	49.94	62.76
	IM21k-SP							80.28	51.02	64.03
	RingMo	0.06	0.07	17071	71.24	13.9	81.29	80.09	49.24	53.17
	MTP	0.04	0.04	37400	85.91	10.0	84.75	76.81	49.05	58.18
AIRS	IM1k-SP	3.81	4.39	20614	84.99	21.9	104.78	*81.60*	**52.88**	**67.36**
	IM21k-SP							**82.19**	**53.59**	**68.25**
	RingMo	3.81	4.39	20286	84.66	17.5	102.34	**81.90**	51.48	*64.56*
	MTP	4.24	4.44	39366	90.43	11.5	97.46	*80.31*	51.07	63.57
Our Method										
TEA	IM1k-SP	4.71	5.43	11106	45.79	16.2	77.51	81.41	52.84	66.30
	IM21k-SP							81.61	52.58	67.51
	RingMo	4.71	5.43	11117	46.40	13.3	77.78	*80.96*	51.23	64.32
	MTP	4.98	5.21	18800	43.19	10.5	88.98	80.31	*51.21*	*64.17*
	IM1k-SP	7.41	8.54	*12484*	*51.47*	16.7	79.90	81.77	52.32	67.02
	IM21k-SP							*82.08*	52.60	67.45
	RingMo	7.41	8.54	*12488*	*52.12*	13.6	79.53	80.32	*51.34*	65.20
	MTP	7.82	8.19	*20332*	*46.71*	10.7	90.68	80.16	51.73	64.83
	IM1k-SP	10.73	12.37	13873	57.20	18.0	86.12	**81.85**	*52.61*	*66.94*
	IM21k-SP							81.95	52.76	68.11
	RingMo	10.73	12.37	13868	57.88	14.1	82.46	80.82	**51.73**	**65.33**
	MTP	11.32	11.85	22130	50.84	11.0	93.22	**81.45**	**51.78**	**65.76**

orientated object detection models with a limited training budget.

b) TEA generalizes well across different pretraining pipelines: The results in Table IV are obtained

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TABLE VI

RESULTS OF SCENE CLASSIFICATION ON UCM, AID, AND NWPU-RESISC45. "↑" INDICATES THE HIGHER THE BETTER. "↓" INDICATES THE LOWER THE BETTER. WE TEST THE TRAINING MEMORY FOOTPRINT AND THE TRAINING TIME OF EACH METHOD ON THE SAME MACHINE AND REPORT THE TEST RESULTS ON NWPU-RESISC. "IM1K-SP" MEANS THE EXTRACTOR IS SUPERVISED PRETRAINED ON IMAGENET-1K AND "IM21K-SP" MEANS THE EXTRACTOR IS PRETRAINED ON IMAGENET-21K. FOR TEA AND PEFT METHODS, WE EMPLOY **"BOLD"** FONT TO HIGHLIGHT THE TOP METRIC, "UNDERLINE" FOR THE NEXT BEST METRIC, AND *"Italics"* FOR THE THIRD-RANKING METRIC. DIFFERENT COLORS ARE APPLIED TO HIGHLIGHT DIFFERENT EXTRACTORS, WITH THE MAPPING TABLE AS FOLLOWS: {IM1K-SP, IM21K-SP, RINGMO, MTP}

Methods	Pre-trained	Trainable Params		Training				Dataset (Top1-Acc) ↑		
				Memory		Time Cost		UCM	AID	NWPU-RESISC45
		# (M) ↓	% ↓	MB ↓	% ↓	h ↓	% ↓			
Traditional Tuning Methods										
FULL	IM1k-SP	86.75	100.00	11119	100.00	1.0	100.00	99.05	96.79	95.04
	IM21k-SP							99.05	97.29	95.50
	RingMo	86.75	100.00	12136	100.00	1.0	100.00	96.76	95.00	94.06
	MTP	95.53	100.00	8066	100.00	0.8	100	98.10	95.55	94.03
HEAD	IM1k-SP	0.00	0.00	1423	12.80	0.68	68.00	86.48	78.28	75.11
	IM21k-SP							89.05	80.66	78.84
	RingMo	0.00	0.00	1457	12.01	0.5	50.00	13.81	11.91	23.49
	MTP	0.00	0.00	876	10.86	0.3	37.50	70.10	78.64	75.74
PEFT Methods										
SIDE	IM1k-SP	3.73	4.30	**2371**	**21.32**	0.8	80.00	98.19	95.68	93.82
	IM21k-SP							98.10	95.85	94.12
	RingMo	3.73	4.30	**2366**	**19.50**	0.7	70.00	97.33	95.63	93.89
	MTP	4.14	4.33	**2500**	**30.99**	0.6	75.00	97.71	94.58	92.87
ADAPTER	IM1k-SP	3.52	4.06	7517	67.61	0.9	90.00	98.76	96.36	95.00
	IM21k-SP							98.86	96.78	95.05
	RingMo	3.52	4.06	8213	67.67	0.8	80.00	95.43	92.74	89.57
	MTP	3.94	4.12	5471	67.83	0.7	87.50	90.95	89.89	87.33
LORA	IM1k-SP	3.50	4.04	7314	65.78	1.4	140.00	98.86	96.56	94.26
	IM21k-SP							99.14	96.99	94.45
	RingMo	3.50	4.04	8186	67.45	1.0	100.00	50.67	44.20	56.30
	MTP	3.95	4.13	5348	66.30	0.8	100.00	72.95	80.86	79.77
BITFIT	IM1k-SP	0.21	0.24	7029	63.22	0.9	90.00	94.00	91.63	88.44
	IM21k-SP							95.62	93.54	90.48
	RingMo	0.21	0.24	7706	63.50	0.8	80.00	42.57	39.59	53.04
	MTP	0.11	0.13	5092	63.13	0.6	75.00	74.95	80.83	81.23
LN	IM1k-SP	0.06	0.07	7026	63.19	0.9	90.00	95.14	91.66	88.24
	IM21k-SP							95.71	94.15	94.10
	RingMo	0.06	0.07	7706	63.50	0.8	80.00	38.67	43.95	63.65
	MTP	0.04	0.04	5089	63.09	0.6	75.00	71.14	79.90	79.77
AIRS	IM1k-SP	3.81	4.39	8752	78.71	1.2	120.00	**98.95**	**96.93**	**95.15**
	IM21k-SP							**99.33**	**97.51**	**95.38**
	RingMo	3.81	4.39	9391	77.38	1.1	110.00	94.38	93.45	90.40
	MTP	4.24	4.44	6202	76.89	0.8	100.00	93.24	92.30	88.58
Our Method										
TEA	IM1k-SP	4.71	5.43	3919	35.25	0.9	90.00	98.86	96.38	94.24
	IM21k-SP							*99.05*	97.08	*94.97*
	RingMo	4.71	5.43	3583	29.52	0.7	70.00	98.00	*95.79*	*94.23*
	MTP	4.98	5.21	3746	46.44	0.8	80.00	*98.00*	*95.25*	*93.57*
	IM1k-SP	7.41	8.54	*4647*	*41.79*	0.9	90.00	*98.86*	96.39	94.06
	IM21k-SP							98.76	*97.04*	94.95
	RingMo	7.41	8.54	*4313*	*35.54*	0.8	80.00	**98.19**	95.89	94.48
	MTP	7.82	8.19	*4387*	*54.39*	0.6	75.00	**98.38**	95.40	**94.28**
	IM1k-SP	10.73	12.37	5418	48.73	1.0	100.00	98.67	*96.49*	*94.39*
	IM21k-SP							98.67	97.03	94.90
	RingMo	10.73	12.37	5041	41.54	0.8	80.00	*97.90*	**96.38**	**94.54**
	MTP	11.32	11.85	5020	62.24	0.7	87.50	98.19	**96.00**	94.27

based on a self-supervised pretrained model (RingMo) and show that TEA performs pretty well. Thus, the effectiveness of TEA is independent of the pretraining pipeline.

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3) Semantic Segmentation: To further verify the effectiveness of TEA on dense prediction tasks, we conduct experiments on three semantic segmentation datasets, including Vaihingen, LoveDA, and iSAID. The results are reported in Table V. We also adopt Swin-B [38] models pretrained on ImageNet-1k and ImageNet-21k as the off-the-shelf feature extractor. RingMo and MTP are also adopted to evaluate TEA.

a) TEA performs competitively to FULL and surpasses most PEFT methods with a better tradeoff between training efficiency and performance: According to the results in Table V, we can see that TEA achieves the comparable performance to FULL on iSAID with 42% memory usage reductions and more than 13% training time savings. On Vaihingen and LoveDA, TEA performs slightly inferior to FULL but surpasses most PEFT methods with better training efficiency.

b) TEA shows a more significant training time advantage, when dealing with semantic segmentation tasks: The results in Table V show that TEA achieves a more significant training time advantage compared to FULL with a 28% training time reduction with 5.43% trainable parameters.

4) Scene Classification: We also test TEA on classic scene classification datasets. According to the results in Table VI, we see that TEA can also achieve advanced performance among the PEFT methods. The gap between the tested methods is relatively smaller than that of other tasks. It can be attributed to the fact that the scene classification task is relatively simple and the metrics of these datasets reach a near saturation point. However, TEA still performs better than most PEFT methods with a better tradeoff between training efficiency and performance.

C. Scalability and Versatility

To evaluate TEA further, we study the scalability and versatility of TEA.

We adjust the scale of the feature extractor to study the scalability of TEA. Specifically, we turn to Swin-L and conduct experiments on DIOR to explore the generalization ability across the VFM volume of TEA. The experimental settings remain the same as Section IV-B. The results are shown in Table VII. We can see that TEA can still achieve advanced performance and the improvement of training efficiency brought by TEA is much more significant with a larger feature extractor. It indicates that TEA can be easily scaled up to larger models. The parameter efficiency of TEA is also becoming more obvious with the model size increasing, as the size of our SIDEAN is fixed. Although TEA outperforms FULL, the gap between TEA and AIRS is relatively larger than that of Swin-B. We speculate that it is because the feature dimensionality of our SIDEAN, e.g., the GhostnetV2, is much smaller compared with that of Swin-L. Mapping between the high-dimensional and low-dimensional feature spaces leads to a larger information loss. We will explore this problem in the future.

To explore the versatility of TEA, we conduct extra experiments on DIOR with ConvNeXt [39] as the feature extractor. ConvNeXt is a modern CNN and can perform comparably to the sophisticated ViTs. We adopt ConvNeXt-B with the weights pretrained on ImageNet as our feature extractor.

TABLE VII

STUDIES OF SCALABILITY AND VERSATILITY, AND RESULTS OF LIGHTWEIGHT GHOSTNETV2. "SWIN-L" IS ADOPTED FOR THE SCALABILITY STUDY AND "CONVNEXT-B" IS ADOPTED FOR THE VERSATILITY STUDY. "↑" INDICATES THE HIGHER THE BETTER. "↓" INDICATES THE LOWER THE BETTER. WE TEST THE TRAINING EFFICIENCY METRICS ON A SINGLE A40 FOR "SWIN-L" AND ON A SINGLE "3080TI" FOR "CONVNEXT". "SWIN-L" IS SUPERVISED PRETRAINED ON IMAGENET-21K WHILE "CONVNEXT" IS PRETRAINED ON IMAGENET-1K. FOR TEA AND PEFT METHODS, WE EMPLOY **"BOLD"** FONT TO HIGHLIGHT THE TOP METRIC, "UNDERLINE" FOR THE NEXT BEST METRIC, AND *"Italics"* FOR THE THIRD-RANKING METRIC

Methods	Backbone	Trainable Params		Training				Dataset (mAP) ↑
				Memory		Time Cost		
		# (M) ↓	% ↓	MB ↓	% ↓	h ↓	% ↓	DIOR
Traditional Tuning Methods								
FULL	Swin-L	195.00	100.00	19077	100.00	17.3	100.00	76.59
	ConvNeXt-B	87.57	100.00	11326	100.00	10.5	100.00	74.25
HEAD	Swin-L	0.00	0.00	4038	21.17	7.5	43.35	69.93
	ConvNeXt-B	0.00	0.00	3528	31.15	4.6	43.81	63.56
PEFT Methods								
SIDE	Swin-L	3.86	1.98	**5835**	**30.59**	12.2	70.52	74.31
	ConvNeXt-B	3.73	4.26	**5265**	**46.49**	7.0	66.67	68.97
ADAPTER	Swin-L	7.90	4.05	13300	69.72	15.3	88.44	77.01
	ConvNeXt-B	5.28	6.03	8436	74.48	8.5	80.95	71.65
LORA	Swin-L	7.92	4.06	14808	77.62	21.6	124.86	76.11
	ConvNeXt-B	9.19	10.50	7342	64.82	8.3	79.05	68.19
BITFIT	Swin-L	0.31	0.16	12635	66.23	15.0	86.71	72.14
	ConvNeXt-B	0.13	0.15	7235	63.88	8.0	76.19	68.70
LN	Swin-L	0.10	0.05	12622	66.16	14.8	85.55	70.93
	ConvNeXt-B	0.04	0.05	7106	62.74	7.0	66.67	68.43
AIRS	Swin-L	8.48	4.35	14693	77.02	19.6	113.29	**77.83**
	ConvNeXt-B	5.71	6.52	9875	87.19	12.5	119.05	73.02
Our Method								
TEA	Swin-L	5.29	2.71	6826	35.78	12.0	69.36	76.34
	ConvNeXt-B	4.71	5.38	6027	53.21	8.0	76.19	*73.40*
	Swin-L	8.16	4.19	*7423*	*38.91*	12.2	70.52	*76.64*
	ConvNeXt-B	7.41	8.46	*6613*	*58.39*	10.5	100.00	74.15
	Swin-L	11.66	5.98	8044	42.17	14.0	80.92	76.45
	ConvNeXt-B	10.73	12.25	6684	59.01	11.3	107.62	**74.60**
GhostNetV2								
FULL	GhostNetV2-10	3.49	-	4604	-	7.0	-	67.16
	GhostNetV2-13	5.82	-	5101	-	7.1	-	70.47
	GhostNetV2-16	8.75	-	5584	-	7.5	-	70.84

The experimental settings remain the same as Section IV-B, except for the re-adjustment of the learning rate. The results are shown in Table VII. We can see that TEA can still achieve superior performance with ConvNeXt, while other PEFT methods see a performance drop. It indicates that TEA is a versatile method that can be easily applied to feature extractors with different architectures.

D. Comparison With Lightweight Methods

We also compare TEA with lightweight methods to highlight the significant performance gains of TEA. We adopt the lightweight GhostNetV2 [63] as the feature extractor and conduct experiments on DIOR. The results are shown in Table VII. According to the results, we can see that the lightweight models can achieve much better training efficiency, while TEA equipped with the same scale GhostNetV2 achieves a much better performance. We can say that TEA bridges the lightweight networks and FMs and achieves a better tradeoff between training efficiency and performance.

E. Performance Comparison With SoTA Remote Sensing FMs

We compare the effectiveness of TEA against that of other well-established remote sensing FMs. To elaborate,

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TABLE VIII

PERFORMANCE COMPARISON OF METHODS ON DIOR. WE EMPLOY **"BOLD"** FONT TO HIGHLIGHT THE TOP METRIC, "UNDERLINE" FOR THE NEXT BEST METRIC, AND *"Italics"* FOR THE THIRD-RANKING METRIC. WE GIVE LESS EMPHASIS TO THE SKYSENSE (SWIN-H), GIVEN ITS CONSIDERABLY LARGER PARAMETER COUNT IN COMPARISON TO THE OTHER MODELS. "SWIN-IMP" REFERS TO SWINTRANSFORMER PRETRAINED ON IMAGENET-21K. "–" MEANS THE CORRESPONDING METRIC IS NOT REPORTED IN THE PUBLISHED ARTICLE

Model	Backbone	# Trainable Params (M)	Memory (MB)	DIOR (mAP)
GASSL [50]	ResNet-50	24	–	67.40
SatMAE [56]	ViT-L	307	–	70.89
RingMo [17]	Swin-B	87	–	75.90
RVSA [20]	ViT-B	86	–	73.22
TOV [54]	ResNet-50	24	–	70.16
CMID [53]	Swin-B	87	–	75.11
CACo [58]	ResNet-50	24	–	66.91
SatLas [55]	Swin-B	87	–	74.10
GFM [52]	Swin-B	87	–	72.84
Scale-MAE [59]	ViT-L	307	–	73.81
SkySense [49]	Swin-L	197	–	76.74
	Swin-H	654	–	78.73
Swin-IMP	Swin-L	197	20023	76.95
TEA	Swin-L	**5**	**8122**	76.82
	Swin-L	8	8719	*76.87*
	Swin-L	*12*	*9348*	**76.96**

we compare the TEA tuning performance of Swin-L, which is pretrained on ImageNet21k, with the full fine-tuning results of SoTA remote sensing FMs on DIOR. The results in Table VIII for this comparison are gathered from their official publications or as reported in SkySense [49]. We align the task framework settings to SkySense with FasterRCNN for object detection on the DIOR dataset. The experimental results indicate that TEA manages to deliver performance that is competitive to that of established SoTA models such as SkySense, and it does so with notable efficiency in both the number of trainable parameters and the training cost.

F. Ablation Studies

We conduct extensive ablation studies to discuss the key designs of TEA. We use Swin-B as the feature extractor and conduct experiments on DIOR. The base experimental settings remain the same as Section IV-B. And the ablation experiment setup is shown in Fig. 7.

1) Choosing a Good SIDEAN Can Boost the Performance: Generally, any visual network can serve as the SIDEAN, while we've found that thoughtfully designed visual blocks can enhance TEA's performance. We have tried three SIDEAN settings, e.g., residual blocks [94], MNASNet [64], and GhostnetV2 [63]. For the setting of residual blocks, we tailor the dimensions of the ResNet modules to (1/8) of the corresponding VFM layers and synchronize the layer counts between SIDEAN and VFM. We list the results in Table IX.

According to the results, we can see that stacking residual blocks parallel to pretrained layers provides ordinary performance with more parameters. MNASNet also performs not that satisfactorily while GhostNetv2 shows better results. We think that modern powerful lightweight networks can balance the parameter volume and model capability better while keeping the advanced training efficiency. Moreover, initializing the SIDEAN with pretrained weights will boost the performance further as shown in Table IX, and we adopt the pretrained SIDEAN for all experiments.

TABLE IX

CASE STUDIES ON THE ARCHITECTURE OF SIDEAN. WE USE SWIN-BASE AS THE FEATURE EXTRACTOR AND CONDUCT EXPERIMENTS ON DIOR. THE BASE EXPERIMENTAL SETTINGS REMAIN THE SAME AS SECTION IV-B. "W/WO PT" MEANS WITH OR WITHOUT LOADING PRETRAINED WEIGHTS. "MNET" REFERS TO "MNASNET", AND "GNETV2" REFERS TO "GHOSTNETV2"

Methods	SIDEAN	Trainable Params		Training		Dataset (mAP) ↑
		# (M) ↓	% ↓	Memory MB ↓	Time Cost h ↓	DIOR
Our Method						
TEA	Residual Blocks	6.65	7.67	7247	12.3	72.91
	MNet-13 (wo PT)	4.92	5.67	6216	9.83	72.67
	MNet-13 (w PT)					72.99
	GNetV2-10 (wo PT)	4.71	5.43	**6036**	11.0	73.41
	GNetV2-10 (w PT)					**74.02**

TABLE X

CASE STUDIES OF OUT INDICES AND GUIDANCE SETTINGS. WE SHOW EACH CASE BY SCHEMATICS AND REPORT CORRESPONDING DIOR RESULTS. "OI" INDICATES OUT INDICES, AND "DIAG" REFERS TO DIAGRAM. "LG" REFERS TO LAYER GUIDANCE, WHILE "SG" REFERS TO STAGE GUIDANCE

OI	Diag	DIOR (mAP) ↑
EARLY	a-1	73.67
LAST	a-2	73.73
UNIFORM	a-3	74.02

(a)

LG	SG	DIOR (mAP)↑		iSAID (mIoU)↑	
		Swin-B	RingMo	Swin-B	RingMo
✗	✗	73.83	74.27	64.96	63.39
✗	✓	74.00	74.52	65.58	63.54
✓	✗	73.98	74.62	65.14	63.56
✓	✓	**74.02**	**74.90**	**66.30**	**64.32**

(b)

2) Selection of VFM Out Indices Will Affect TEA's Performance: As we mentioned in Section III-C, we adopt the LayerDrop strategy and have to determine the output layers of the VFM. It is worth noting that we mainly conduct the selection strategies on "Stage3" of SwinTransformer, as the blocks of SIDEAN correspond closely to the pretrained layers in other stages. We experiment with three selection strategies, the "early," "last," and "uniform" in Table X. The "early" indicates that we choose the first several layers in "Stage3," while the "last" means we adopt the last blocks. The "uniform" means that we select the layers uniformly. The results show that "early" and "last" lead to inferior performance, and "uniform" provides a satisfactory result. Thus we adopt "uniform" as our default setting.

3) Top-Down Guidance Plays an Important Role: We observe that the pretrained VFMs have learned rich general visual representations and the features close to the top are already semantic-aware. Thus we propose to guide the training of the SIDEAN with the high-level features. We conduct experiments with/without the top-down guidance. Specifically, we employ Swin-B and RingMo to evaluate the influence of guidance across various FMs. Furthermore, we extend our investigation to include both object detection and semantic segmentation tasks on the DIOR and iSAID datasets to

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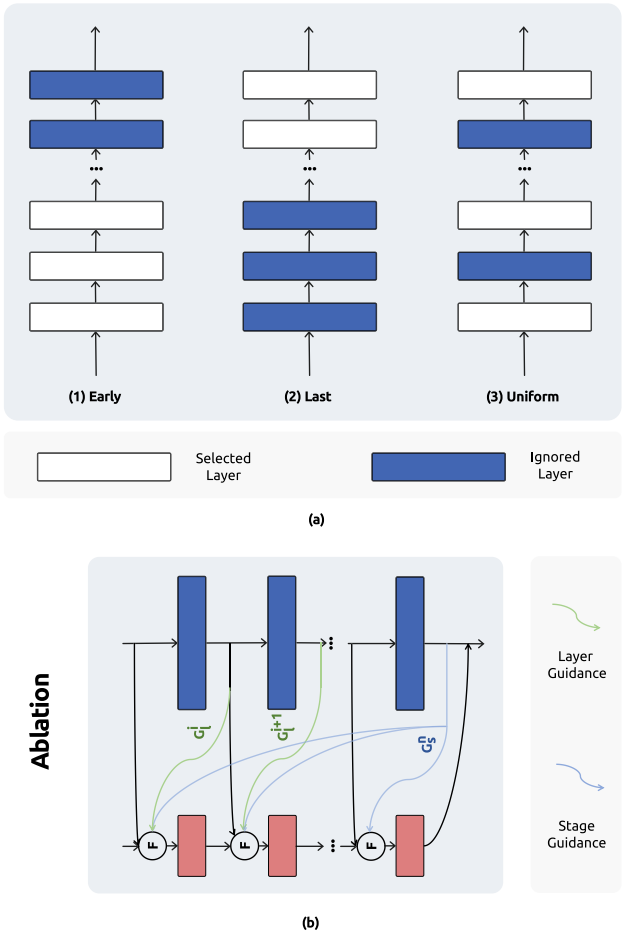


Fig. 7. Ablation studies. (a) Layer selection. We ablate output features of different layers of the off-the-shelf VFMs. "Early" means we select the first several layers, while "last" means the opposite. "Uniform" indicates that we select the layers uniformly. (b) Guidance strategies. We try to drop the layer and stage guidance separately or together.

explore the varying requirements for guidance in different tasks. We list the results in Table X. The results reveal nuances in the efficacy of top-down guidance depending on the task and FM at hand. Nonetheless, a consistent pattern emerges, both forms of guidance play a role in safeguarding the performance of TEA, with the combined application of both demonstrating optimal results on a range of tested FMs and tasks. Consequently, we consider the top-down guidance strategy to be helpful.

V. Conclusion

In the era of FMs, fine-tuning serves as an effective technique in adapting large pretrained models to downstream tasks. In our work, we observe that the traditional full fine-tuning may not be the best choice for tuning pretrained VFMs to perform remote sensing tasks with a limited resource budget, as it is not always training-efficient. We propose a TEA framework to pursue better training efficiency while keeping or boosting performance. We conduct extensive experiments on multiple remote sensing datasets with a number of FMs and demonstrate the effectiveness of TEA. On all the test datasets, the memory consumption of TEA can be compressed over 50% of that of the full fine-tuning, and better performance can be achieved by increasing the memory budget. For dense prediction tasks, TEA can reduce the training time by about 15%. In summary, TEA can serve as an alternative to fine-tuning for adapting FMs to remote sensing tasks, which can achieve an impressive tradeoff between training efficiency and task performance. Moreover, TEA is a general framework that is VFMs-agnostic and can be applied to many VFMs. Also, it is task-agnostic and SIDEAN-agnostic, which ensures its wide applicability. In conclusion, TEA is potentially a promising framework for adapting FMs for remote sensing applications. In future work, we will try to explore the performance upper bound of TEA and extend it to more remote sensing tasks.

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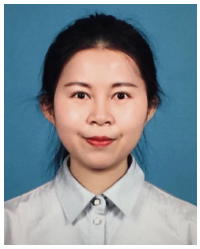
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